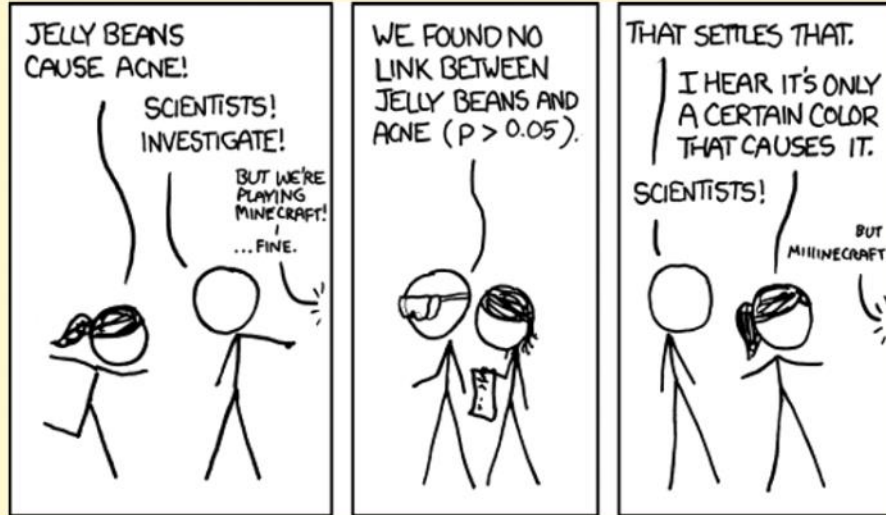


CV 5101
Modeling, Uncertainty, and
Data for Engineers
(July – Nov 2026)

Dr. Prakash S Badal

L-13 (HT)
Sep 25, 2026

xkcd

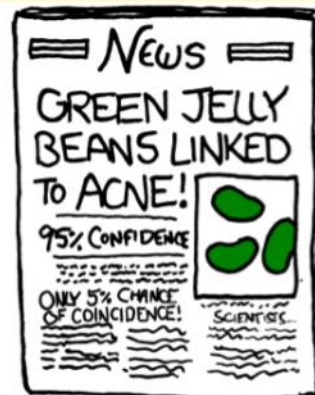


<https://xkcd.com/882>



<https://xkcd.com/882>

<https://explainxkcd.com/882>

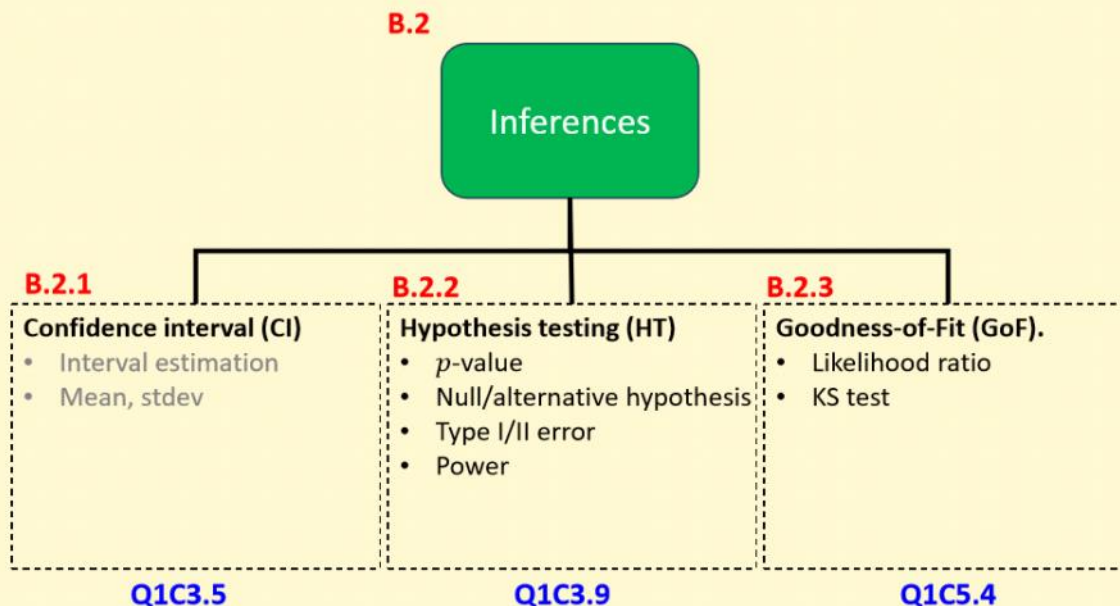


Parameter estimation

- Next remaining classes:
 - Estimation: MoM, MLE
 - Inference: Confidence interval (CI)
 - **Inference**: Hypothesis testing, p -value, power
 - **Goodness of fit (GoF)**: χ^2 , KS
- Risk & reliability (Q2C8)
 - Introduction
 - Risk Evaluation
- **Extreme value analysis (Q2C7)**
 - Peak over threshold
 - Return period and design life

5

Module Overview



Flow

- Null/alternative hypothesis
- Type I/II error
- Hypothesis testing method
- Error calculations
- Power of test
- p -value
- GoF

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Central limit theorem: recap

- For large sample size n , **sample mean** approaches

a **normal distribution**,

with **mean μ** (same as population mean), and

variance σ^2/n (less than population Var.).

Spreadsheet demo!

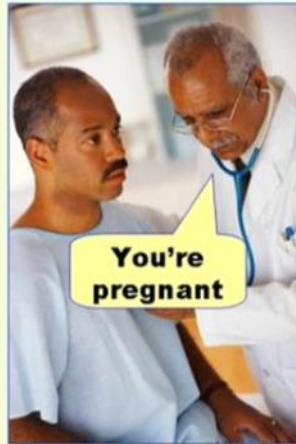
8

Spot the error*

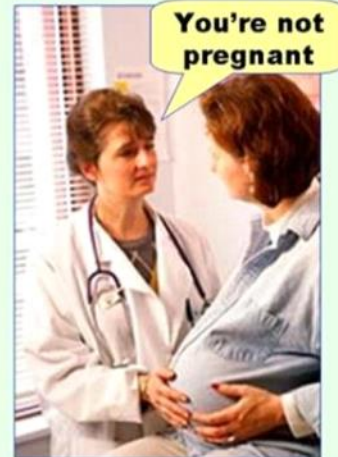
*: not an example from real life

Accuracy of the test
false positive
false negative
Result of the test

Type I error
(false positive)



Type II error
(false negative)



People are NOT generally pregnant.
NULL HYPOTHESIS
 H_0 : YOU ARE NOT PREGNANT → default
 H_1 (ALTERNATIVE): YOU ARE PREGNANT.

9

Justice system in a democracy

- How can someone be arrested if they really are **presumed innocent**?
- Why is a defendant pronounced "**Not Guilty**" instead of innocent?
- Why do citizens put up with a system that allows **criminals to go free on technicalities**?

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Common challenges in justice, research (life?)

→ **Strong Urge** to believe in an unusual events:

e.g. * A new medicine cures cancer.

* A charge person (aka 'defendant') is guilty

→ **Serious Repurcussions** of being wrong

e.g. * Two types of wrong decisions are possible

YOU ACTED ON YOUR URGE, BUT IT TURNED OUT TO BE FALSE	YOU DID NOT ACT ON YOUR URGE, AND IT TURNED OUT TO BE TRUE
→ innocent person goes to jail	→ guilty persons goes free
→ Patient get wrong medicine	→ Patients dying/suffering, despite a cure being available.
→ No Sure shot way to prove the unusual event.	
→ Limited Data	

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Hypothesis testing is the solution!

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Hypothesis testing (& justice system)

No numerical values in courts, but they share four common features:

1. **NULL HYPOTHESIS:** H_0
The presumption of innocence.
2. **ALTERNATE HYPOTHESIS:** H_1
Why a suspected person is arrested.
3. **STANDARD OF JUSTICE:** significance level, α
A reasonable doubt.
4. **A DATA SAMPLE:** Test score.
Evaluation of partial information.

Hypothesis testing (& justice system)

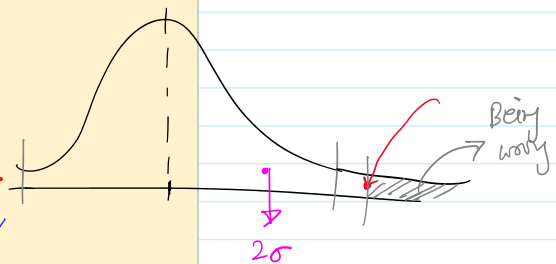
Both statistical testing and the justice system:

1. Concentrate on **rejecting the null hypothesis**

- It is easier
- Rejecting "Presumption of Innocent".
 $\equiv$ "defendant is pronounced guilty"

2. Consider a failure to reject **the null hypothesis**

- As "Not Guilty" verdict
- Proving H_0 is true will take endless evidence.



Hypothesis testing (& justice system)

Neither statistical testing and the justice system are perfect:

- Sometimes, the jury makes an error.
 - An **innocent person goes to jail**
 - Statisticians call it a **Type I**.
- Sometimes, **a guilty person is set free**
 - Statisticians call it a **Type II**.

Which ONE is worse.

Citizens find Type II disturbing but is NOT as horrifying as Type I error.

Type I error = { An Innocent person goes to Jail
+
A guilty person walks free.

L-2/2 (HT)
Sep 30, 2026

Hypothesis testing (& justice system)

No numerical values in courts, but they share four common features:

1. The alternative hypothesis:

This is why a *criminal is arrested* (or a research undertaken)
denoted by H_1 or H_A .

2. The null hypothesis:

The *presumption of innocence* (the solution won't work!)
 H_0 is the logical opposite of H_1 .

3. A standard of justice:

Absolute proof IMPOSSIBLE, hence a *reasonable doubt* (sig. level)
Reject the null hypothesis beyond a reasonable doubt.

4. A data sample:

Evaluation of *partial information* (test score)
The "whole truth and nothing but the truth" is often impossible.

Errors in Hypothesis testing (& justice system)

$$\text{Type I error} \equiv \left\{ \begin{array}{c} \text{an innocent person goes to jail} \\ + \\ \text{a guilty person walks free} \end{array} \right\}$$

HT (& justice system) puts a lot of emphasis on
avoiding Type I error.

Counter example: Airport security

Innocent person and alarm goes off (Type I error)

Guilty person and alarm goes not go off (Type II error)

Sometimes, Type II error is worse!

e.g., airport security.

Errors in Hypothesis testing (& justice system)

- An innocent person goes to jail *Type I*
- A guilty person is set free *Type II*

Errors in Hypothesis testing (& justice system)

HT & justice lady put a lot of emphasis on
avoiding Type I error.

Product example:

- Null hypothesis, H_0 : a product satisfies customer requirements.
- If H_0 is rejected, do not sell the product to customers.

H_0 : New iPhone is substantially improved.

Type I error: Reject an improved model by mistake.
A very expensive error.

Type II error: Failing to reject a bad model & shipping it to customers.
Loss of reputation

Which one is worse? *Type II error.*

Spot the error type

According to a report from the United States Environmental Protection Agency, burning one gallon of gasoline typically emits about 8.9 kg of CO₂. A fuel company wants to test a new type of gasoline designed to have lower CO₂ emissions. Here are their hypotheses:

$$H_0 : \mu = 8.9 \text{ kg}$$

$$H_a : \mu < 8.9 \text{ kg}$$

(where μ is the mean amount of CO₂ emitted by burning one gallon of this new gasoline).

Under which of the following conditions would the company commit a Type I error?

Choose 1 answer:

- ☐ A The mean amount of CO₂ emitted by the new fuel is actually 8.9 kg, and they fail to conclude it is lower than 8.9 kg.
- ☐ B The mean amount of CO₂ emitted by the new fuel is actually lower than 8.9 kg, and they conclude it is lower than 8.9 kg.
- ☐ C The mean amount of CO₂ emitted by the new fuel is actually 8.9 kg, and they conclude it is lower than 8.9 kg.
- ☐ D The mean amount of CO₂ emitted by the new fuel is actually lower than 8.9 kg, and they fail to conclude it is lower than 8.9 kg.

A large university is curious if they should build another cafeteria. They plan to survey a sample of their students to see if there is strong evidence that the proportion interested in a meal plan is higher than 40%, in which case they will consider building a new cafeteria.

Let p represent the proportion of students interested in a meal plan. Here are the hypotheses they'll use:

$$H_0 : p \leq 0.40$$

$$H_a : p > 0.40$$

What would be the consequence of a Type II error in this context?

Choose 1 answer:

- ☐ A They don't consider building a new cafeteria when they should.
- ☐ B They don't consider building a new cafeteria when they shouldn't.
- ☐ C They consider building a new cafeteria when they shouldn't.
- ☐ D They consider building a new cafeteria when they should.

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Spot the error type

A quality control expert wants to test the null hypothesis that a new solar panel is no more effective than the older model.

What would be the consequence of a Type I error in this context?

Choose 1 answer:

- ☐ A They don't conclude the new panel is more effective when it actually is not more effective.
- ☐ B They don't conclude the new panel is more effective when it actually is more effective.
- ☐ C They conclude the new panel is more effective when it actually is more effective.
- ☐ D They conclude the new panel is more effective when it actually is not more effective.

According to a report from the United States Environmental Protection Agency, burning one gallon of gasoline typically emits about 8.9 kg of CO₂. A fuel company wants to test a new type of gasoline designed to have lower CO₂ emissions. Here are their hypotheses:

$$H_0 : \mu = 8.9 \text{ kg}$$

$$H_a : \mu < 8.9 \text{ kg}$$

(where μ is the mean amount of CO₂ emitted by burning one gallon of this new gasoline).

Which of the following would be a Type II error in this setting?

Choose 1 answer:

- ☐ A The mean amount of CO₂ emitted by the new fuel is actually 8.9 kg, and they conclude it is lower than 8.9 kg.
- ☐ B The mean amount of CO₂ emitted by the new fuel is actually lower than 8.9 kg, and they fail to conclude it is lower than 8.9 kg.
- ☐ C The mean amount of CO₂ emitted by the new fuel is actually 8.9 kg, and they fail to conclude it is lower than 8.9 kg.
- ☐ D The mean amount of CO₂ emitted by the new fuel is actually lower than 8.9 kg, and they conclude it is lower than 8.9 kg.

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Example 1

- Consider bars made of IS 1786-compliant **Fe500 grade steel**
 - Let's focus on the mean yield strength
 - We are interested in deciding **whether or not** the mean yield strength is 500 MPa

Formally,

$$H_0: f_y = 500 \text{ MPa}$$

$$H_1: f_y \neq 500 \text{ MPa}$$

Four situations may arise

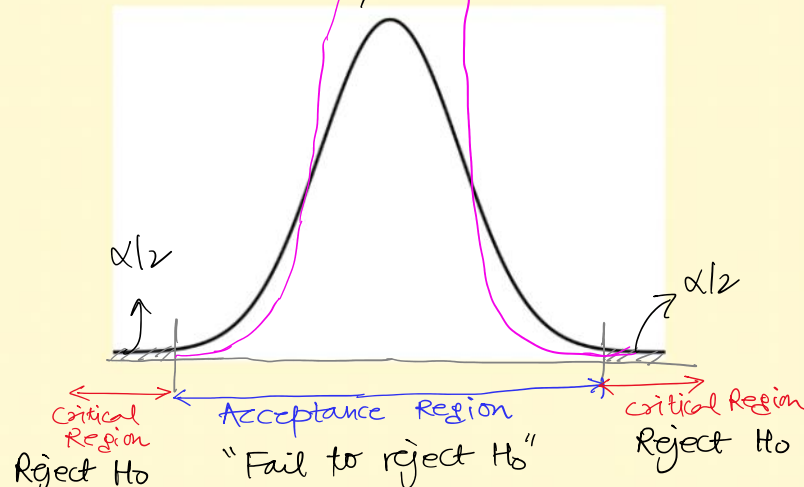
Decision ↓	H_0 is True	H_0 is False
Fail to reject H_0	No error	Type II error (β)
Reject H_0	Type I error (α)	No error

Example 1

significance level: α

$$H_0: \mu_{F_y} = 500 \text{ MPa}$$

$$H_1: \mu_{F_y} \neq 500 \text{ MPa}$$



Type I error

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Probability of Type I Error, α

- $\alpha = \text{Pr}(\text{Type I error})$
= $\text{Pr}(\text{reject } H_0 \text{ when } H_0 \text{ is True})$

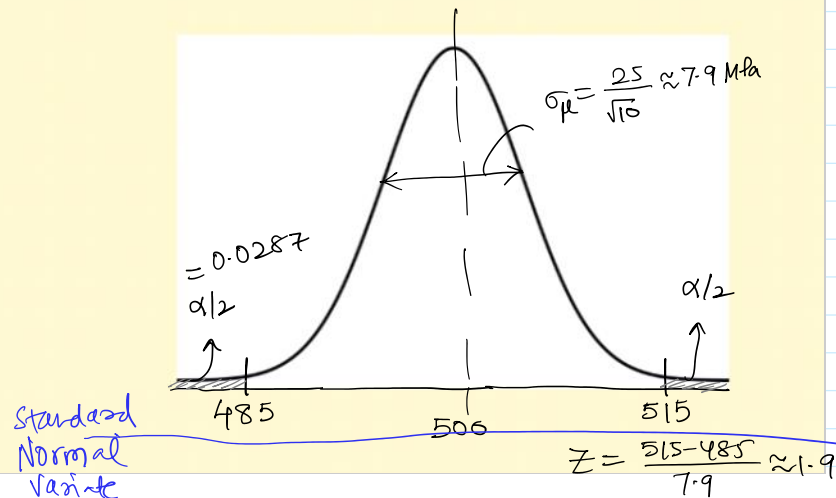
We can reduce Type I error

- by widening the acceptance region
- by increasing the number of samples

Probability of Type I Error, α

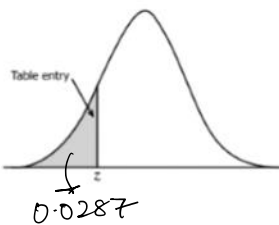
- Consider Fe500 bar again; suppose $\sigma_{Fe500} = 25 \text{ MPa}$
- Suppose we tested a sample with $n = 10$ specimens
- If a client accepts bars with average $f_y \in [485 \text{ MPa}, 515 \text{ MPa}]$
- Find the **probability of Type I error**?

$$\alpha = 5.74\%$$



Standard Normal Probabilities

Z-table



z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
-3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
-3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
-3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
-3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
-2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
-2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
-2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
-2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
-2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
-2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
-2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
-2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
-2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
-2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
-1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
-1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
-1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
-1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
-1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
-1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
-1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
-1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
-1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
-1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
-0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
-0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
-0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
-0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
-0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
-0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
-0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
-0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
-0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
-0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641

Table entry for z is the area under the standard normal curve to the left of z .

Type II error

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Probability of Type II Error, β

- $\alpha = \text{Pr}(\text{Type I error}) = \text{Pr}(\text{Reject } H_0 \text{ when } H_0 \text{ is True})$
- $\beta = \text{Pr}(\text{Type II error})$
 $= \text{Pr}(\text{failure to reject } H_0 \text{ when } H_0 \text{ is False})$

To calculate β ,
we need a "specific alternative hypothesis."

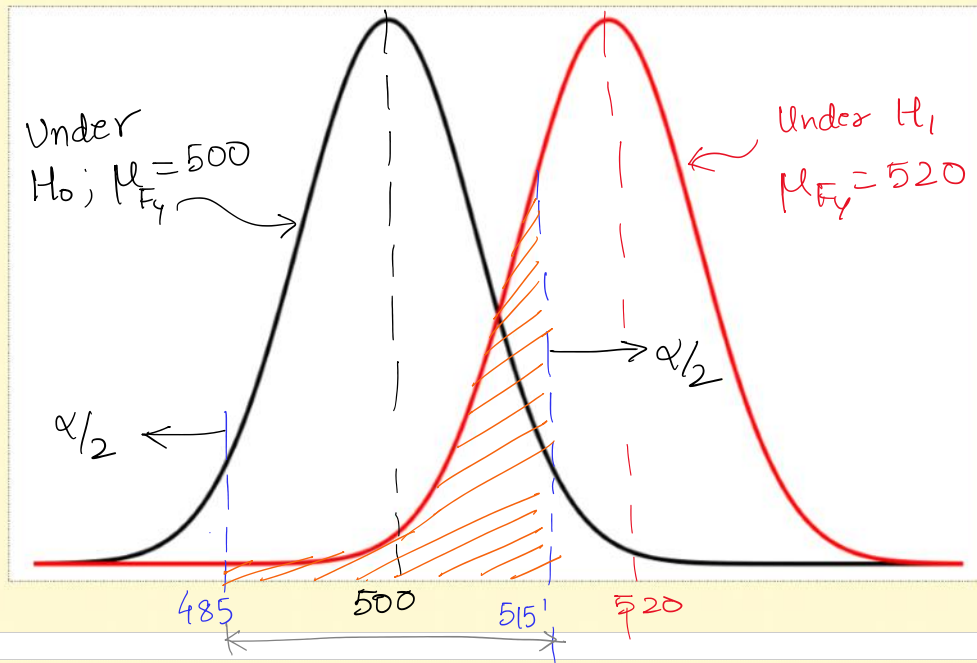
e.g.

$$H_0: \mu_{F_y} = 500 \text{ MPa}$$

versus

$$H_1: \mu_{F_y} = 520 \text{ MPa}$$

Type II error



Probability of Type II Error, β

- Fe500; $\sigma_{Fe500} = 25 \text{ Mpa}$; we tested a sample with $n = 10$ specimens
- If an alternative hypothesis exists that $\mu_{F_y} = 520 \text{ MPa}$
- Find the **probability of Type II error?**

$\beta = \text{Pr}(\text{failure to reject } H_0 \text{ when } H_1 \text{ is True})$

$$= \text{Pr}(\bar{X}_{F_y} \in [485, 515] \mid \mu_{F_y} = 520)$$

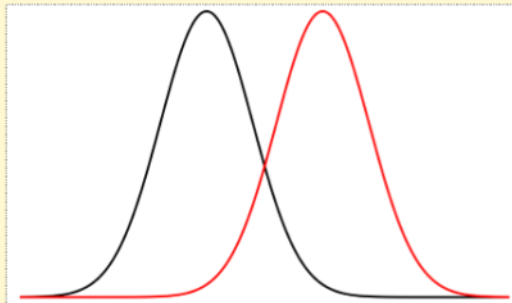
$$= \text{Pr}(\bar{X}_{F_y} \leq 515 \mid \mu_{F_y} = 520) - \text{Pr}(\bar{X}_{F_y} \leq 485 \mid \mu_{F_y} = 520)$$

$$= \Phi(Z_1) - \Phi(Z_2)$$

Where $Z_1 \approx \frac{515 - 520}{7.9} \approx -0.63$

$$Z_2 \approx \frac{485 - 520}{7.9} \approx -4.43$$

$$\Rightarrow 0.2643 - 0.0000 \approx 0.2643$$



Power of a statistical test

- The **power** of a statistical test

$\Pr(\text{correctly rejecting } H_0 \text{ when } H_1 \text{ is True})$

$$\text{power} = 1 - \beta$$

Four situations may arise

Decision ↓	H_0 is True	H_0 is False
Fail to reject H_0	No error	Type II error (β)
Reject H_0	Type I error (α)	No error $\equiv \text{power} = 1 - \beta$

Power: steel strength example

- In the steel strength problem, we found that $\beta = 0.2643$ when the true mean is $\mu = 520$, what is the power of this test

$$\text{Power} = 1 - \beta \approx 73.6\%$$

p-value

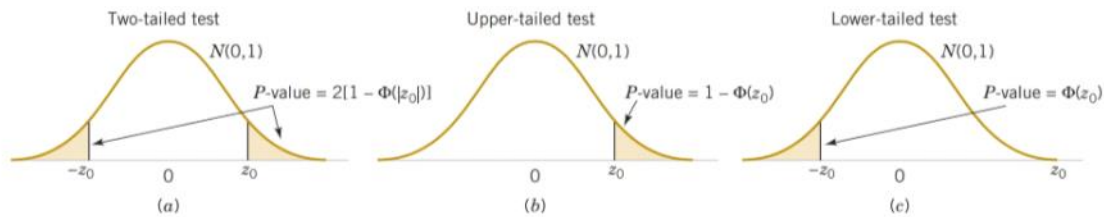
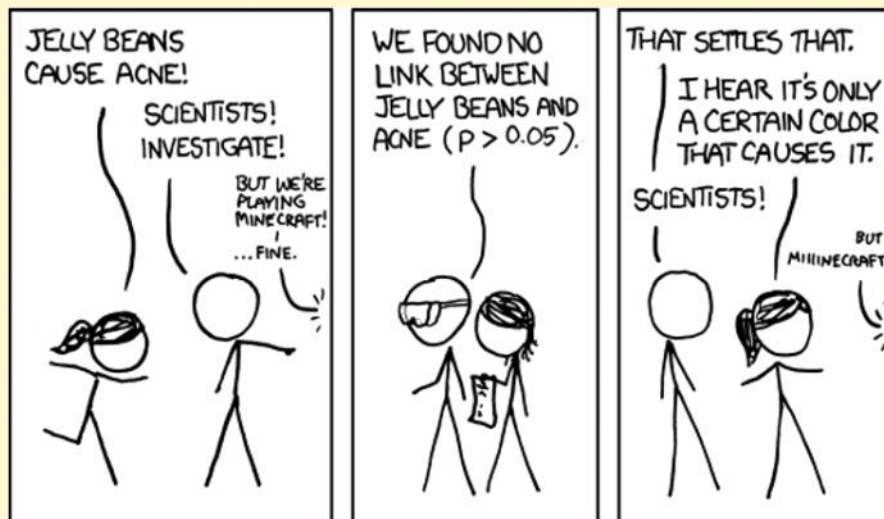
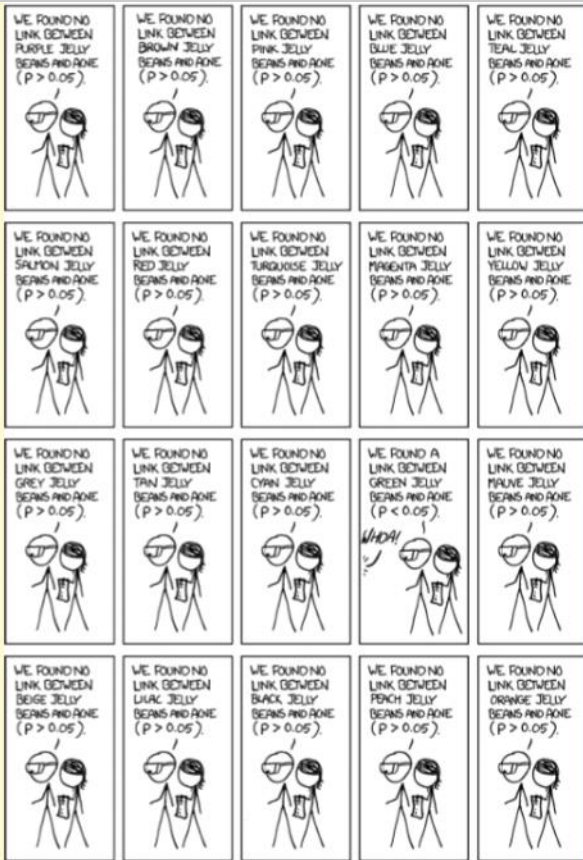


FIGURE 9-10 The P -value for a z -test. (a) The two-sided alternative $H_1: \mu \neq \mu_0$. (b) The one-sided alternative $H_1: \mu > \mu_0$. (c) The one-sided alternative $H_1: \mu < \mu_0$.

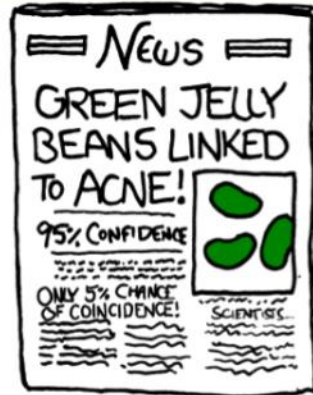
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<https://xkcd.com/882>



<https://xkcd.com/882>
<https://explainxkcd.com/882>



Questions, comments,
or concerns?